

# Practical Guidelines for AI-Assisted Historical Interpretation

## Introduction

My first experience with generative AI was humbling in an unexpected way.

Curious to see what this remarkable new technology knew about me, I asked it about my own career. It confidently informed me that I had died in 2017.

At the time, that response was more amusing than concerning. Large language models were still in their infancy. They often mixed fact with fiction, confidently filled gaps in their knowledge, and occasionally invented entire biographies. The experience reinforced what many historians already believed: AI was an interesting curiosity, but certainly not something to trust with historical work.

How quickly that changed.

In only a few short years, generative AI has evolved from producing obvious mistakes to becoming a genuinely useful research and interpretation tool. The change has been so rapid that almost any paper describing today's software risks becoming outdated before it is published. Models improve. Interfaces change. New capabilities appear every few months. Workflows that seemed impossible one year become commonplace the next.

That rapid evolution presents a challenge for anyone attempting to document the field. I do not present these guidelines as best practices because the field is evolving far too quickly for anyone to

make that claim. They are simply the practices that proved successful during one project at one moment in the evolution of AI-assisted historical interpretation.

This paper is therefore not intended to be a manual for any particular AI system, nor a collection of prompts that will inevitably become obsolete. Instead, it is offered as a set of field notes from one historian's journey across a bridge that many museums, historians, educators, and researchers are only now beginning to cross.

The observations that follow were learned while developing *The Prescott Girls* project and exploring how generative AI might contribute to historically responsible interpretation. The project became a laboratory in which ideas could be tested, discarded, refined, and tested again. Some approaches proved remarkably effective. Others failed completely. Many of the most useful lessons were discovered only after making the same mistakes repeatedly.

It is also important to recognize that *The Prescott Girls* is not the final destination. It has served as the research environment in which these methods were developed. The more meaningful test still lies ahead: applying AI-assisted historical interpretation within the Pownalborough Court House, where its success will ultimately be measured not by the quality of the images themselves, but by whether they help visitors better understand the lives of the people who once lived there.

The purpose of these guidelines is not to define a single "correct" workflow. Better methods will undoubtedly emerge as the technology continues to mature. Rather, they are offered in the hope that others beginning this journey can benefit from the lessons already learned, avoid some of the same dead ends, and spend more of their time on what matters most: careful research, thoughtful interpretation, and meaningful engagement with history.

The bridge between traditional historical practice and AI-assisted interpretation is still under construction. These pages are simply field notes from one traveler who has begun the crossing.

# 1. Start With Research, Not Prompts

If there is one lesson that underlies every other guideline in this paper, it is this: AI is not a substitute for historical research. In fact, the opposite is true. The better the historical research, the more useful AI becomes.

One of the earliest surprises in this project was discovering how directly the quality of an image reflected the quality of the underlying historical understanding. When I began with only a general idea, "a nineteenth-century kitchen" or "a girl sewing by the fire", the results often looked plausible at first glance but quickly revealed historical inaccuracies. Clothing drifted toward the wrong decade. Architectural details borrowed from other regions. Furniture, tools, and household objects reflected statistical averages rather than the specific time and place I was trying to recreate.

As the historical research became more detailed, those problems diminished. The AI was no longer inventing the world; it was helping to visualize one that had already been carefully reconstructed from historical evidence.

Before generating a single image, I found it worthwhile to establish as much of the historical context as possible. Depending on the project, that might include:

- The time period and geographic location.
- The architecture of the buildings.
- Clothing appropriate to the season, region, and social setting.
- Household objects, tools, and furnishings.
- Daily activities and occupations.
- Family relationships and social customs.

The research does not eliminate interpretation; it establishes its boundaries. Just as a museum diorama or an artist's reconstruction is constrained by the available evidence, AI-assisted interpretation should remain grounded in what is known while being transparent about what is inferred.

This preparation also changes the nature of the work. Instead of spending hours correcting historically implausible images, those same hours can be devoted to exploring alternative interpretations, refining compositions, and asking more interesting historical questions. The conversation shifts from "Can the AI produce the right image?" to "Which of these historically plausible interpretations best communicates the story we want visitors to understand?"

That distinction is subtle but important. AI should not replace historical scholarship. It should amplify it.

## **2. Build the Scene Around a Story**

One of the most important lessons I learned had nothing to do with AI at all. It had to do with storytelling.

Early in the project I spent considerable time trying to create beautiful historical portraits. The results were often attractive, historically plausible, and technically impressive, but they rarely held my attention for very long. They documented what someone looked like, but they didn't invite me to wonder who that person was or what was happening in their life.

Everything changed when I stopped asking AI to create portraits and began asking it to illustrate moments.

Museums tell stories. Every artifact, document, and room exists because something happened there. The goal of an interpretation should therefore be to help visitors imagine those moments. Rather than

asking, "Can you show me a girl in an 1830s dress?" I found it much more productive to ask, "What might Sallie have been doing while her mother prepared supper?" or "How would the family have reacted when a long-awaited letter arrived from Philadelphia?"

Those questions naturally led to scenes that felt alive.

Throughout *The Prescott Girls*, many of the strongest interpretations grew from ordinary moments of everyday life:

- Sallie stealing a fingerful of pie filling while helping in the kitchen.
- Uncle Thomas reading beside the fire in his swing-chair.
- The Prescott family making apple cider together.
- The girls preparing the house for visiting relatives.
- The family gathered around a newly arrived letter from Philadelphia.

None of these moments are dramatic in the traditional sense. Their strength comes from their familiarity. They invite visitors to recognize themselves in people who lived nearly two centuries ago.

When an interpretation succeeds, the visitor's first response should not be, "That's a beautiful picture."

It should be:

*"What's happening here?"*

That simple question encourages curiosity, and curiosity leads naturally to historical engagement. Visitors begin looking more closely at the clothing, the architecture, the objects on the table, and the relationships between the people. The image becomes more than an illustration; it becomes the beginning of a conversation.

For museums, that distinction is important. AI should not simply produce attractive images. It should help create interpretations that encourage visitors to ask questions, imagine daily life, and form a deeper connection with the people whose stories we are trying to preserve.

### **3. Develop Characters Before Building Scenes**

One of the greatest challenges in any historical illustration is consistency. Visitors quickly recognize when the same individual appears different from one image to the next, even if they cannot immediately identify why. Hair changes. Facial proportions shift. Clothing evolves unexpectedly. The result is a subtle loss of continuity that can distract from the historical story.

I discovered this problem almost immediately while developing *The Prescott Girls*. My first instinct was to create complete narrative scenes. Although many of the individual images were successful, Beckie, Louisa, and Sallie often emerged looking like different children from one illustration to the next. Each image solved the immediate problem, but together they failed to establish recognizable historical characters.

The solution was to reverse the process.

Before attempting family scenes or museum interpretations, I began creating individual character studies. Each study focused on a single person rather than a narrative. The objective was not to tell a story, but to establish a visual identity that could be carried consistently throughout the project.

For each principal character, I worked to define:

- Facial features
- Hair style and color
- Body proportions
- Height relative to other family members

- Facial expressions
- Clothing appropriate to age and season
- Appearance as the child matured over time

These studies became a reference library that informed every subsequent illustration. Instead of asking the AI to invent Beckie or Louisa each time, I could begin with a well-established visual foundation and concentrate on the story unfolding around them.

This approach produced another unexpected benefit. Once the characters became familiar, I found myself thinking less about creating images and more about portraying people. Decisions about posture, expressions, and interactions became easier because the girls had developed distinct personalities. Historical interpretation became less about illustration and more about visual storytelling.

The same principle applies beyond people. Buildings, furniture, household objects, and landscapes also benefit from developing consistent reference studies before they are incorporated into larger scenes. Establishing these visual anchors early creates continuity throughout a project and allows future effort to focus on interpretation rather than reconstruction.

Character studies may seem like a detour from creating finished illustrations, but they are an investment. Time spent establishing consistency at the beginning is repaid many times over as a project grows in scope and complexity.

## **4. Develop Historical Clothing in Layers**

Historical clothing proved to be one of the more challenging aspects of the project. Early attempts to generate a fully dressed nineteenth-century figure often produced garments that blended details from

different decades, omitted important layers, or introduced modern elements that immediately broke the historical illusion.

Over time, I found it much more effective to approach clothing as it was actually constructed.

Rather than asking AI to produce a complete outfit in a single step, I first established the figure itself. Once the proportions and appearance were satisfactory, I added the appropriate undergarments and preserved that version as a reference. Dresses, aprons, shawls, coats, bonnets, and accessories were then added incrementally, with each successful stage becoming the starting point for the next.

This approach mirrors the way historical clothing was worn and, perhaps not surprisingly, often produces more convincing results. It also makes individual corrections much easier. If an outer garment requires revision, the underlying clothing and figure remain intact instead of having to recreate the entire character from the beginning.

The process may appear slower, but it generally reduces the amount of rework required later. Like many aspects of AI-assisted interpretation, investing additional effort early in the workflow often saves considerably more time during refinement.

## **5. Expect Complexity to Compound**

One of the more surprising lessons from this project was how quickly complexity increases as a scene grows. The relationship is not linear. Each additional person, building, or significant object introduces new interactions that the AI must maintain simultaneously. What begins as a straightforward illustration can quickly become a challenging exercise in balancing dozens of visual relationships.

This phenomenon reminded me of software development. A small program may be easy to understand, but every new feature creates additional interactions with the features already in place.



Eventually the complexity comes not from the individual components, but from the relationships between them. AI-assisted image generation behaves in much the same way.

A single historical figure is relatively straightforward. Add a second person and the AI must maintain believable proportions, eye contact, body language, and lighting between them. Add a third and those relationships multiply. Introduce a historic building, furniture, or an animal, and the composition becomes more demanding still.

During this project I found the following rule of thumb to be useful.

| <b>Scene Element</b>  |  | <b>Relative Complexity</b>                        |
|-----------------------|--|---------------------------------------------------|
| One person            |  | Low                                               |
| Two people            |  | Moderate                                          |
| Three people          |  | Manageable                                        |
| Four people           |  | Significant                                       |
| Five people           |  | Difficult                                         |
| Distinct architecture |  | Often comparable to adding another character      |
| Animals               |  | Frequently comparable to adding another character |

For example, an illustration of the three Prescott girls standing before the Pownalborough Court House is effectively a four-subject composition. The building is not simply a background. It has its own architectural details, perspective, scale, lighting, and historical accuracy that must be maintained alongside the people.

Likewise, a family scene that includes four people, the court house, and a horse may contain six major subjects competing for the model's attention. As complexity increases, the likelihood of unexpected errors also increases. A corrected face may alter a hand. Improved clothing may distort proportions. Architectural details may drift while the figures improve.

Fortunately, this is one area where rapid advances in AI are already becoming apparent. Models continue to improve in their ability to maintain consistency across increasingly complex scenes. Even so, understanding how complexity compounds remains valuable because it helps set realistic expectations and encourages breaking difficult interpretations into manageable stages rather than attempting everything at once.

Whenever possible, begin with the simplest version of a scene that successfully tells the historical story. Complexity should support the interpretation, not become the focus of the work.

## **6. Review the Entire Image After Every Revision**

One of the most surprising characteristics of AI-assisted image generation is that corrections are rarely isolated. Improving one part of an image often changes something else that previously appeared correct. Over time, I came to think of this as "global drift." The entire image remains in motion, even when I am attempting to change only a single detail.

A typical editing session might begin by correcting a dress that belongs to the wrong decade. The revised clothing is historically accurate, but now the character's face has subtly changed. Correcting the face may alter the position of the hands. Repairing the hands might shift the perspective of the room or introduce inconsistencies in the lighting. Solving one problem often reveals another.

Early in the project I made the mistake of examining only the area I had intended to change. That proved to be an inefficient way to work because significant errors often appeared elsewhere in the

image without immediately attracting attention. Eventually I adopted a different habit. After every meaningful revision, I stopped looking at the corrected detail and instead reviewed the entire composition as though I were seeing it for the first time.

That simple change in workflow prevented many small errors from accumulating into larger ones.

As the project progressed, I developed a mental checklist that I used before considering an illustration complete.

## **People**

- Are the characters the correct ages?
- Are facial features consistent with earlier illustrations?
- Do expressions match the story being told?
- Are hands and fingers anatomically correct?
- Do body positions appear natural?
- Are the characters looking where they should?

## **Clothing**

- Is the clothing appropriate for the period?
- Are all necessary layers present?
- Do fabrics and construction remain consistent?
- Does the clothing fit naturally?
- Have any modern details appeared?

## **Architecture**

- Does the building match the historical evidence?

- Are rooflines, windows, and doors correct?
- Are proportions consistent throughout the structure?
- Are materials appropriate for the period?

## **Objects**

- Are furnishings and tools historically appropriate?
- Are objects correctly scaled?
- Do they appear in sensible locations?
- Has anything modern appeared unexpectedly?

## **Environment**

- Does the landscape match the geography?
- Is the vegetation appropriate?
- Is the lighting consistent?
- Does the weather support the scene?

## **Story**

- Is the intended action immediately understandable?
- Does every element support the historical narrative?
- Would a museum visitor understand what is happening without additional explanation?

The final questions are often the most important. A technically perfect illustration has limited value if it does not communicate the historical story. Every decision should ultimately serve the visitor's understanding rather than the technology itself.

## 7. Know When to Start Over

Persistence is usually a virtue in historical research. It is not always a virtue in AI-assisted image generation.

Occasionally an illustration reaches a point where every correction seems to create a new problem. A revised dress changes the character's posture. Correcting the posture alters the perspective. Repairing the perspective affects the architecture. After several rounds of editing, the image may actually be farther from the goal than it was when the process began.

At first, I assumed that continuing to refine the image would eventually solve these problems. Experience taught me otherwise. Sometimes the AI has simply moved into an unproductive direction, and no amount of additional editing will recover the original intent.

Learning to recognize that moment is an important skill.

When progress begins to stall, I have found it helpful to stop, step away from the image, and begin again with a fresh generation. Because the historical research, character studies, and reference material have already been established, starting over is often much faster than attempting to rescue an increasingly unstable composition.

There is also a natural tendency to fall victim to the sunk cost fallacy. After investing an hour refining an image, it becomes psychologically difficult to abandon it, even when it is clear that progress has stalled. I found myself thinking, "I've already invested so much time in this version." In reality, that previous effort is gone regardless of what I do next. The only meaningful question is whether the next hour is more likely to produce a better result by continuing to edit the current image or by beginning a new one.

More often than I expected, starting over proved to be the faster path.

## 8. Understand Safety Systems

One aspect of AI-assisted historical interpretation that surprised me was how frequently modern safety systems intersected with legitimate historical work.

Most image-generation systems include safeguards designed to prevent the creation of harmful or inappropriate content. These safeguards are an important part of responsible AI development, particularly when children are involved. Museum professionals should expect to encounter them and understand that they exist for good reasons.

Historical interpretation, however, often depicts activities that were entirely ordinary in the past but may appear unusual when viewed through a modern lens. Children carrying water from a well, climbing into a hayloft, gathering firewood, working beside adults, or using simple hand tools were common parts of daily life in the nineteenth century. An AI system may recognize these situations as potentially risky before it recognizes their historical context.

When this occurred, I found that the solution was rarely to argue with the system. Instead, it was usually more productive to communicate the historical setting more clearly. Additional context, describing the broader scene, emphasizing normal family activities, or breaking a complex interpretation into smaller steps often helped the AI better understand the intended purpose.

Patience also proved valuable. A request that failed one day might succeed with a clearer description or as newer models became available. Because these systems continue to evolve rapidly, some limitations disappear while new ones inevitably emerge.

The broader lesson extends beyond safety systems themselves. AI responds best when it understands not only what we want to create, but why we are creating it. Historical interpretation

depends on context. The more faithfully we communicate that context, the more successful the resulting interpretation is likely to be.

Responsible use of AI means respecting both the historical evidence and the safeguards built into the technology. Neither should be viewed as an obstacle. Both exist to encourage thoughtful, responsible interpretation.

## **9. Historical Interpretation Is an Iterative Process**

One of the earliest assumptions I brought to AI-assisted illustration was that success meant producing the "right" image. It did not take long to realize that this was the wrong way to think about the process.

Historically informed interpretation is almost never the result of a single successful generation. It emerges through repeated cycles of research, experimentation, evaluation, and refinement. Each iteration becomes an opportunity to compare competing ideas, discover historical inconsistencies, and ask whether the story can be communicated more effectively.

Throughout *The Prescott Girls* project, nearly every significant illustration evolved over time. Clothing changed as additional research clarified period details. Architecture was revised as new photographs and measured drawings became available. Furniture, household tools, room layouts, and building details were all revisited repeatedly as my understanding of the historical evidence deepened.

The process came to resemble art direction more than image generation. Every improvement resulted from collaboration. Historians corrected clothing. Museum professionals corrected architecture. Needleworkers corrected construction details. AI generated alternatives. The final interpretations emerged from the conversation among all of them.

One of the most significant changes AI brings to historical interpretation is that it changes the economics of iteration. Traditionally, each commissioned illustration represented a substantial investment of both time and money. Museums and publishers often accepted the strongest interpretation they could reasonably afford because every additional revision increased the cost.

Generative AI changes that equation.

Because alternative interpretations can now be explored quickly and economically, it becomes practical to compare many possibilities before selecting one. The value is not simply that AI produces images faster. The value is that historians can spend more time thinking. They can reject weaker ideas, revisit the evidence, refine details, and continue asking whether a better interpretation exists.

In many respects, this mirrors the way historians already work. We revise manuscripts. We reconsider conclusions when new evidence appears. We refine museum exhibits after observing how visitors respond. AI simply extends that same process of thoughtful refinement into the visual interpretation of history.

The first image is rarely the destination. It is the beginning of the conversation.

## **10. Maintain a Reference Library**

One of the most valuable habits to emerge from this project was preserving successful work, even when it was not part of a finished illustration.

Early in the process I tended to save only completed scenes. Over time I realized that many of the intermediate studies were actually more valuable. A successful character study, an accurately reconstructed dress, a carefully researched fireplace, or a historically correct room could be reused repeatedly in future interpretations.



Rather than viewing these images as individual results, I began treating them as a growing reference library.

Depending on the needs of a project, that library may include:

- Character studies
- Wardrobe studies
- Building studies
- Furniture and household object studies
- Landscape studies
- Architectural details
- Lighting studies

These references become the visual equivalent of research notes. They preserve successful solutions, reduce the need to solve the same problem repeatedly, and improve consistency across an entire body of work. As the historical understanding evolves, individual reference studies can also be updated without requiring every completed illustration to be recreated from the beginning.

This approach mirrors traditional museum practice. Museums routinely maintain reference photographs, measured drawings, conservation records, artifact catalogs, and architectural surveys. AI-assisted interpretation benefits from the same discipline. The reference library simply becomes another research collection, one that supports visual interpretation rather than replacing historical evidence.

I also found that these studies often became useful in ways I had not anticipated. A building created for one illustration might later serve as the background for another. A well-developed character study could be reused years later as the foundation for a completely different scene. Each successful study became another resource available for future work.

The more comprehensive the reference library became, the less time I spent recreating what I had already learned, and the more time I could devote to exploring new historical questions. Like any good research collection, its value grew with every addition.

## **11. The Goal Is Interpretation, Not Perfection**

Throughout this paper I have described techniques for creating AI-assisted historical illustrations. It is worth concluding by remembering why museums create visual interpretations in the first place.

Museums have always relied on visual storytelling to help visitors understand worlds they cannot directly experience. Paintings, dioramas, architectural reconstructions, artist renderings, and scale models all serve the same purpose. They help transform historical evidence into something visitors can see, imagine, and better understand.

AI-assisted imagery belongs within that tradition.

Its purpose is not to recreate the past with photographic certainty. Such certainty is rarely possible. Every historical interpretation contains elements that are firmly documented alongside others that represent informed judgment based on the available evidence.

The goal is therefore not perfection.

The goal is to help visitors visualize a historically informed interpretation that encourages curiosity, supports learning, and creates a deeper appreciation for the people who once lived these lives.

Like every other interpretive tool available to museums, AI should be judged not by the sophistication of the technology, but by how effectively it helps tell the historical story.

An illustration succeeds when visitors stop looking at the technology and begin thinking about the history.

## 12. Transparency Matters

The introduction to this paper described these guidelines as field notes from an ongoing journey. That description is intentional because AI-assisted historical interpretation continues to evolve rapidly. New capabilities will emerge, new workflows will develop, and many of the techniques described here will undoubtedly improve over time.

The principles of responsible interpretation, however, are unlikely to change.

Visitors deserve to understand what they are seeing.

Whenever practical, museums should distinguish historical photographs from AI-assisted interpretations, explain which elements are supported directly by historical evidence and which represent informed interpretation, cite historical sources when they are available, and be honest about uncertainty when the historical record is incomplete.

Transparency does not diminish the value of an interpretation. It strengthens it. Visitors appreciate understanding how historians reconstruct the past and where evidence ends and informed judgment begins.

One final lesson deserves mention.

Do not fall in love with an image too early.

The first image that feels magical is often the one that contains the most subtle historical errors. During this project, many of my favorite illustrations were eventually discarded or substantially revised because additional research revealed inaccuracies that I had initially overlooked.

That experience reflects another familiar lesson from historical research. The first explanation is rarely the final one. Better evidence leads to better interpretation.

The same is true here.

The image that ultimately serves a museum best may not be the first version or even the most visually striking. More often, it is the fifth or tenth iteration after careful research, thoughtful revision, and repeated comparison with the historical evidence.

AI makes those additional iterations practical. Historical scholarship makes them meaningful.

If there is a single idea I hope readers take away from these field notes, it is this: artificial intelligence is not replacing historical interpretation. AI has not diminished the importance of historical expertise. If anything, it has increased it. The better the historian, the better the interpretation. It is giving historians, museum professionals, educators, and researchers another tool with which to practice it.

The bridge between traditional historical practice and AI-assisted interpretation is still under construction. I hope these observations help make the crossing a little easier for those who follow.

## **About the Author**

Aric Wilmunder is an author, researcher, and former software engineer whose work explores the intersection of historical research, museum interpretation, and emerging technologies. The methods described in this paper were developed during the ongoing *The Prescott Girls* project and continue to evolve through collaboration with historians, museums, and educators.

# About This Research

## The Prescott Girls Historical Research Series

*Practical Guidelines for AI-Assisted Historical Interpretation* is part of an ongoing effort to document the people, artifacts, family connections, historical discoveries, and technologies developed that inspired *The Prescott Girls: A Letter from Philadelphia*.

For additional research articles, historical images, schoolgirl samplers, family records, and educational resources, visit:

[www.theprescottgirls.com](http://www.theprescottgirls.com)

A Well-Regulated Press

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